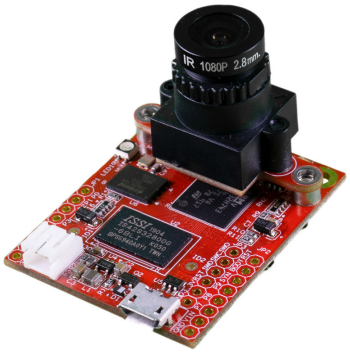
OpenMV® Cam H7 Plus

Datasheet

SKU: OMV-CAM-H7-PLUS-V1 (PCB V4.1)

Description

The OpenMV Cam H7 Plus is an enhanced version of the OpenMV Cam H7, featuring significantly more memory with 32 MB external SDRAM and 32 MB external QUADSPI flash. This allows for larger frame buffers, more complex MicroPython scripts, and higher-resolution image processing. It ships with the OV5640 5 MP autofocus camera module, enabling higher-resolution capture than the standard H7. Built around the same STM32H743II ARM Cortex-M7 processor at 480 MHz.



Target Areas

Machine vision, robotics, industrial automation, IoT, edge AI, TensorFlow Lite, high-resolution imaging, video streaming, barcode/QR scanning.

Key Features

- **High-performance MCU** — STM32H743II ARM Cortex-M7 processor running at 480 MHz with hardware FPU, DSP, and JPEG acceleration.
- **Extended memory** — 32 MB external SDRAM and 32 MB QUADSPI flash for large frame buffers and complex scripts.
- **5 MP camera** — OV5640 autofocus camera module with up to 2592x1944 resolution. M12 lens mount.
- **MicroPython + TensorFlow** — Program vision algorithms in Python with TensorFlow Lite support for neural networks.
- **Rich I/O** — 10 I/O pins with SPI, I2C, UART, CAN, ADC, DAC, PWM, and servo control. All pins are 5V tolerant.
- **Connectivity** — Full-speed USB with virtual COM port and mass storage. microSD card slot for data logging.
- **Compact form factor** — 35.6 mm x 44.5 mm board size with integrated power management and LiPo battery support.

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1 Features and System Components

1.1 Microcontroller

Component	Details
STM32H743II	ARM Cortex-M7 core running at 480 MHz. Features double-precision floating-point unit (FPU), DSP instructions, and hardware JPEG codec. Includes 2 MB on-chip flash and 1 MB SRAM.

1.2 Memory

Component	Details
On-chip Flash	2 MB program/storage flash memory.
On-chip SRAM	1 MB SRAM.
External SDRAM	32 MB SDRAM (32-bit bus, 100 MHz clock, 400 MB/s bandwidth) for large frame buffers and extended heap.
External QUADSPI Flash	32 MB QUADSPI flash (100 MHz, 4-bit DDR mode, 100 MB/s read) for additional program and data storage.
microSD Card	Supports microSD/microSDHC/microSDXC up to 64 GB. 4-bit SDMMC interface at up to 100 Mb/s.

1.3 Connectivity

Component	Details
USB	Full-speed USB 2.0 (12 Mbps) via Micro-USB connector. Enumerates as virtual COM port (CDC) and USB mass storage device (MSC) simultaneously.
SPI	1 SPI bus available on header, up to 80 Mb/s.
I2C	2 I2C buses available (I2C2 on P4/P5, I2C4 on P7/P8). Standard/Fast mode up to 1 Mb/s.
UART	2 UART ports (UART1 on P0/P1, UART3 on P4/P5). Up to 7.5 Mb/s.
CAN	1 CAN bus (CAN2 on P2/P3). Up to 1 Mb/s. Compatible with CAN FD.

1.4 Camera System

Component	Details
Image Sensor	OV5640 1/4-inch 5 MP CMOS image sensor with autofocus. Up to 2592x1944 resolution.
Image Performance	Full 5 MP JPEG capture. For machine vision: up to VGA (640x480) RGB565 or grayscale at high frame rates. TensorFlow Lite inference on captured frames.
Lens	2.8 mm focal length lens on standard M12 lens mount. User-replaceable with any M12 lens.
Camera Interface	8-bit parallel DCMI interface. Removable camera module system — compatible with OV7725, OV2640, MT9V024, MT9V034, and FLIR Lepton modules.

1.5 LEDs

Component	Details
RGB LED	1 user-programmable RGB LED for status indication.
IR LEDs	2 high-power 850 nm infrared LEDs for night vision and IR illumination applications.

1.6 Power Management

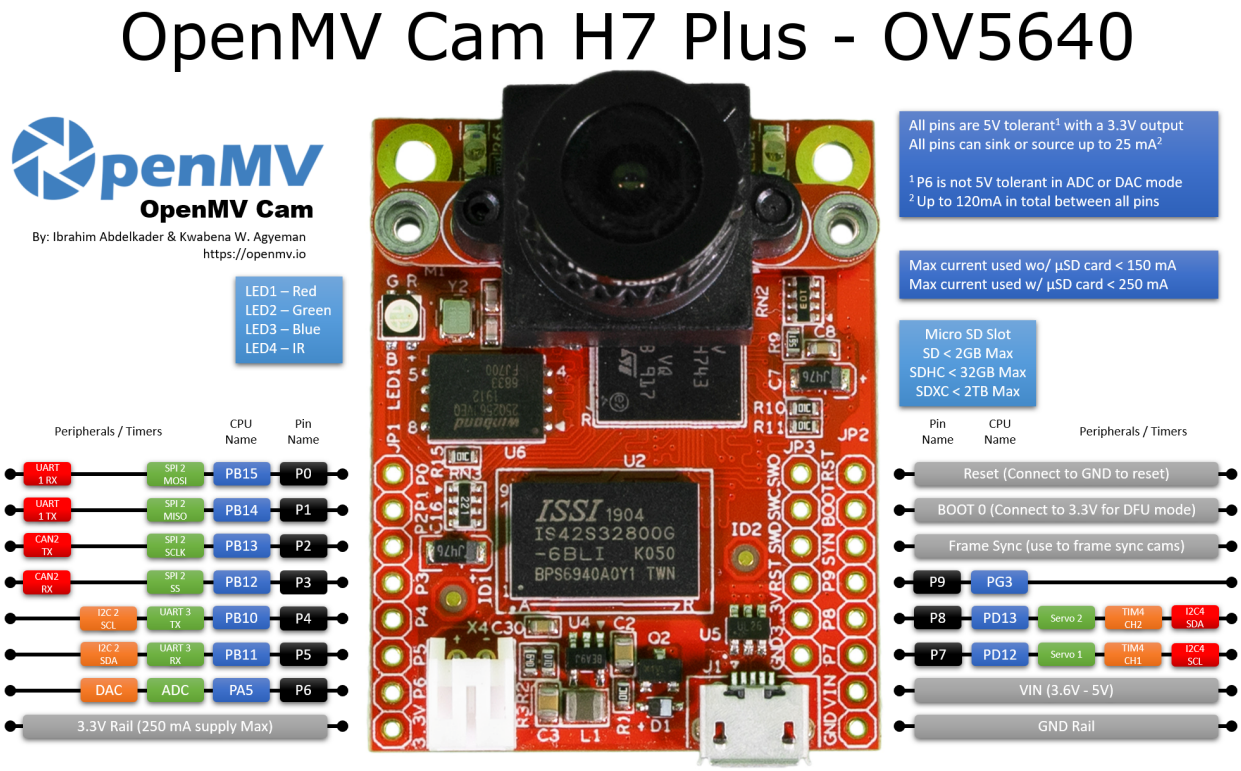
Component	Details
VIN	External power input, 3.6 V to 5.0 V. High-efficiency switching regulator with up to 1 A output.
3.3 V Rail	Regulated 3.3 V output available on header pin. Maximum 250 mA for external loads.
LiPo Battery	JST-SH connector for single-cell 3.7 V LiPo batteries. Integrated charging when USB is connected.
USB Power	Can be powered via Micro-USB.

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VIN	External supply voltage	3.6	5.0	5.0	V
VIO	I/O pin input (digital)	-0.3	—	5.0	V
VIO_A	I/O pin input (ADC/DAC)	-0.3	—	3.6	V
TOP	Operating temperature	-40	25	125	°C

3 Connector Overview



3.1 J1 Header (Main I/O)

Pin	Label	Type	Alternate Functions
1	P0	I/O	UART1 RX / TIM1 CH3N / SPI2 MOSI (PB15)
2	P1	I/O	UART1 TX / TIM1 CH2N / SPI2 MISO (PB14)
3	P2	I/O	CAN2 TX / TIM1 CH1N / SPI2 SCLK (PB13)
4	P3	I/O	CAN2 RX / SPI2 SS (PB12)
5	P4	I/O	TIM2 CH3 / I2C2 SCL / UART3 TX (PB10)
6	P5	I/O	TIM2 CH4 / I2C2 SDA / UART3 RX (PB11)
7	P6	I/O	TIM2 CH1 / DAC / ADC8 (PA5) — 3.3V tolerant only in ADC/DAC mode
8	3V3	Power	3.3 V regulated output (250 mA max)

3.2 J2 Header (Servo / Control)

Pin	Label	Type	Alternate Functions
1	RST	Input	Active-low reset
2	BOOT	Input	Connect to 3.3 V for DFU boot mode
3	SYN	I/O	Frame sync (camera synchronization)
4	P9	I/O	Servo3 / TIM4 CH3 (PD14)
5	P8	I/O	Servo2 / TIM4 CH2 / I2C4 SDA (PD12)
6	P7	I/O	Servo1 / TIM4 CH1 / I2C4 SCL (PD13)
7	VIN	Power	3.6–5.0 V input
8	GND	Power	Ground

3.3 J3 Header (SWD Debug)

Pin	Label	Type	Notes
1	SWC	Debug	Serial Wire Clock
2	SWD	Debug	Serial Wire Data
3	RST	Input	Reset (active low)
4	3V3	Power	3.3 V (500 mA max supply)
5	GND	Power	Ground

4 Functional Overview

4.1 Board Topology

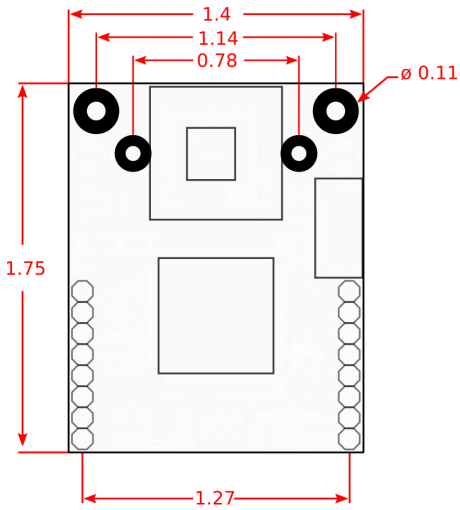
Ref.	Description
U1	STM32H743II microcontroller (ARM Cortex-M7, 480 MHz)
U2	Camera module connector (8-bit parallel DCMI)
U3	Micro-USB connector (Full-Speed USB 2.0)
U4	microSD card slot (4-bit SDMMC)

Ref.	Description
U5	High-efficiency switching voltage regulator (3.3 V, 1 A)
U6	32 MB SDRAM
U7	32 MB QUADSPI flash
LED1	User-programmable RGB LED
LED2/3	850 nm infrared LEDs (x2)
J1	Main I/O header (8 pins)
J2	Servo / control header (8 pins)
J3	SWD debug header (5 pins)
BAT	LiPo battery connector (JST-SH)

4.2 Power Tree

The OpenMV Cam H7 Plus can be powered from three sources: the VIN pin (3.6\u20135.0 V), the Micro-USB connector (5 V bus power), or a single-cell LiPo battery (3.7 V nominal) via the JST-SH connector.

5 General Info



*All dimensions in Inches

Dimension	Value
Board Size	35.6 mm x 44.5 mm (1.4 in x 1.75 in)
Weight	~50 g (with camera module and lens)
Lens Mount	M12 (S-mount), interchangeable
Operating Temperature	-40°C to +85°C
Storage Temperature	-65°C to +150°C

6 Certifications

Certification	Region	Details
RoHS	EU	Compliant with EU Directive 2011/65/EU (RoHS 2) and amendment 2015/863 (RoHS 3). All components and solder are lead-free.
REACH	EU	Compliant with EU Regulation (EC) No 1907/2006.

7 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

8 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/openmv-cam-h7-plus
Documentation	https://docs.openmv.io
Community Forum	https://forums.openmv.io
STM32H743II Datasheet	https://www.st.com/en/microcontrollers/stm32h743ii.html
OV5640 Image Sensor	https://www.ovt.com/products/sensor.php?id=101

9 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.